energy deposition, which prevents a lowering of U_{avg} by radiative transport. Plasma energy density starts to drop only, when the *NA* is reduced below 0.2 and the increase of plasma volume due to the faster upstream movement of the breakdown wave starts to dominate [108,114]. Correspondingly, both plasma energy density and conversion efficiency into bubble energy decrease slowly with decreasing focusing angle for $NA < 0.2$ [114]. Nevertheless, even for $NA = 0.06$, shock wave emission was observed in breakdown by 10-ns, 532-nm pulses and simulations predicted a peak energy density of ≈ 6 kJcm^{-3}, while fs breakdown did not even result in bubble formation [254].

The average plasma pressure values derived from U_{avg} using EOS data lie between 3.6 and 11 GPa (Table 3). They agree well with previous results for ns breakdown at moderate *NA* obtained by measuring the shock wave velocity in the immediate plasma vicinity, which lie between 7 and 12 GPa [40,110,228]. Both methods do not provide peak pressure values within the plasma - the values in hot spots and strings within ns plasmas are certainly much larger. According to the EOS data in Fig. 3, the pressure at 260 kJcm^{-3} energy density (as estimated for the hot spot in the plasma at E = 574 µJ in Fig. 8) should by far exceed 100 GPa (1 Mbar). An experimental validation of this conjecture is still needed. It requires ultrafast time-resolved imaging capable of tracking the formation of shock waves in larger hot spots and their propagation around the thin high-density strings (see section 6.2).

5.5. Bremsstrahlung and blackbody characteristics of femtosecond and nanosecond plasmas

Understanding the bremsstrahlung and blackbody characteristics of plasma luminescence in dependence on laser parameters and plasma properties enables to draw meaningful conclusions on breakdown dynamics and thermodynamic plasma properties. Spatially resolved time-integrated spectral information on fs and ns plasma luminescence in the wavelength range between 410 and 690 nm is presented in Figs. 7 and 8, and quantitative spectroscopic data on spatially integrated luminescence from ps and ns breakdown in water are available in the literature [224-226,246].

There are three sources of plasma radiation involving free-free, free-bound, and bound-bound electronic transitions. The first one involves electron-ion and electron-neutral collisions, in which a photon is emitted. It results in continuum emission and is usually referred to as bremsstrahlung [238,240,289,290,294-296]. The second involves capture of a free electron into a bound state (recombination). It is the inverse of photoionization and results in a continuum below a cut-off wave length, i.e. $\lambda < \lambda_{th}$, where λ_{th} is the photoionization edge for the particular bound state formed. The third radiation process involves electronic transitions between bound states of an atom or ion and results in a

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line spectrum. For hot luminescent high-density plasmas of atoms at low atomic number (z number), bremsstrahlung dominates [294] but the luminescence is often interpreted as blackbody radiation [76,78,224-226,297]. In the spectral range, where radiation can escape the water cell and becomes photographically or spectroscopically detectable, we see merely the long-wavelength tail of the plasma radiation. Here, the blackbody spectrum corresponding to a certain thermodynamic temperature is similar to the thermal bremsstrahlung spectrum at a higher electron temperature [291,295]. Therefore, the luminescence colour on photographs and the shape of the accessible spectrum alone do not allow a clear identification of the nature of the plasma radiation. To facilitate such identification, it is useful to recall that free-free interactions between charged particles always produce bremsstrahlung but this emission assumes the properties of blackbody radiation only under specific conditions [183,291,298]. When the ensemble of CB electrons is in thermal equilibrium, thermal bremsstrahlung is emitted. Equilibrium is not yet reached during optical breakdown by a 100-fs pulse, where the kinetic energy spectrum of CB electrons changes during the pulse and approaches an asymptotic shape only towards its end [149]. However, it will be reached during longer pulses. Part of the bremsstrahlung is always self-absorbed by the plasma, whereby the absorption takes place preferentially at low frequencies because the absorption length equals the inverse-bremsstrahlung decay length [290]. With increasing density of emitting and absorbing particles, the spectrum is self-absorbed up to ever larger frequencies. When all spectral components are self-absorbed (opaque plasma) *and* the particles feature a Maxwell-Boltzmann distribution (thermal equilibrium), we have a blackbody [291]. This condition is fulfilled if the electron density is sufficiently high and the plasma size is sufficiently large, and the pulse is sufficiently long. For a micrometer-sized plasma, the emission of a blackbody-like spectrum requires a supercritical electron density $\rho_c > 10^{21}$ cm^{-3} [236,290,297] but for larger plasmas, subcritical electron densities are sufficient.

We will now use the above criteria to classify the plasma radiation visible on the photographs in Figs. 7 and 8. The plasma radiation produced by 350-fs pulses shown in Fig. 7(a) is likely thermal bremsstrahlung because the electron ensemble will be in thermal equilibrium during most of the pulse. The luminescence colour of the emitted radiation is not only linked to the kinetic energy spectrum of CB electrons evolving during the laser pulse but also to the red-shift and loss of spectral intensity during thermalization [149]. The thermalization process is fast, when the free electron density is low enough to keep the molecular and band structure of water intact. In this case, CB electrons rapidly lose energy

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through inelastic collisions with water molecules and hydrate within less than 300 fs [165-167,170,290]. However, when at higher irradiance and electron density the band structure dissolves and most water molecules dissociate, electron-ion recombination becomes the predominant thermalization pathway, with thermalization times of 20-30 ps [123,158,174]. The change of the thermalization process with increasing electron density will likely influence the spectral emission characteristics of the plasma and is probably responsible for the colour shift from beam waist towards upstream direction in the fs plasma images of Fig. 7(a).

The *total* duration of plasma luminescence is always longer than the laser pulse duration. However, the duration of *local* plasma luminescence may be shorter, when the thermalization time is below τ_L. For ns pulses well above the breakdown threshold, the upstream moving ionization wave shields downstream regions such that thermalization can progress and bremsstrahlung is quenched in its wake. This seems to be the case for ns breakdown at moderate *NA* shown in Fig. 8. Here the downstream region of the plasma between beam waist and hot spot has sharp contours and features a mottled substructure. This indicates that the duration of the luminescence is too short to be blurred by the hydrodynamic plasma expansion leading to bubble formation. The initial bubble wall velocity in ns breakdown is well above 1000 m/s (i.e. 1 µm/ns) for a 1-mJ pulse [40]. Thus, the lack of observable blur indicates a local luminescence duration below 1 ns, which is consistent with thermal bremsstrahlung from CB electrons that ceases during their thermalization. The temperature of the thermalized electron-ion plasma is lower than the initial electron temperature. Its emission is likely weaker such that the colour of the photographs arises mainly from the initial bremsstrahlung. Numerical simulations showed that the CB electron energy spectrum in water is constant over a large irradiance range from below bubble threshold to nearly full ionization of the $1b_1$ valence band level, with average energies of 6.5-7.0 eV [149]. This finding explains the bluish colour and the homogeneity of the luminescence in the downstream plasma region.

The luminescence in and around the hot spot in the upstream plasma region has a very different appearance: it is blurred, whitish, and the colour exhibits a red shift towards the plasma periphery. At E_L = 574 µJ, where the energy density is estimated to be as large as 260 kJ/cm^3, the luminescence extends beyond the laser cone angle. These observations indicate that the plasma emission remains strong, when local thermal equilibrium (LTE) between electron and ion populations is reached and the plasma starts to expand. The duration of plasma luminescence is longer than in the wake of the moving breakdown wave further downstream because laser energy is absorbed during the entire second half of the laser pulse and the resulting thermodynamic temperature is much larger. Due to the large size (10-20 µm) of the

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